Theory

ISPhO

A2-1

English

Vibrational structure of polyynes

Part A. An infinite chain of carbon atoms (6.5 points)

A.1 (0.8pt)

$E =$

A.2 (0.8pt)

$m\ddot{u}_n =$

$m\ddot{v}_n =$

A.3 (0.9pt)

$a =$ $b =$

$c =$ $d =$

A.4 (1.0pt)

$\omega_1/\omega_0 =$

$\omega_2/\omega_0 =$

A.5 (1.0pt)

$\omega_1/\omega_0 \approx$

$\omega_2/\omega_0 \approx$

A.6 (1.0pt)

$\omega_1/\omega_0 \approx$

$\omega_2/\omega_0 \approx$

A.7 (1.0pt)
Plot:

$\alpha = 0.5,\ \beta = 0.1$	$\alpha = 0.5,\ \beta = 3.0$

Part B. Spectroscopy of polyynes (3.5 points)

B.1 (0.8pt)

Equation:

B.2 (0.7pt)

$q_l =$

B.3 (2.0pt)

Single bond:

Triple bond: